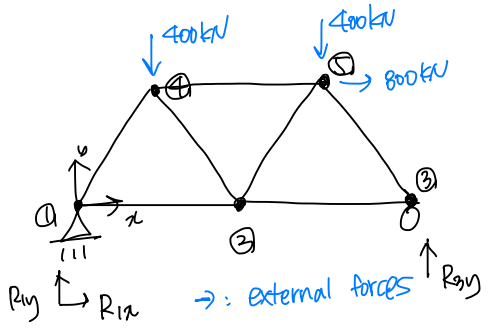
MSA HW week5 Chisung Kim

1. planar truss with 6-10 members



node 1: 0, 0
2: 4000, 0
3: 8000, 0
4: 2000, 3000
5: 6000, 3000

element 1: 1→2
2: 2→3
3: 4→5
4: 1→4
5: 4→2
6: 2→5
7: 5→3

All members has $E = 70GPa$, $A = 4000\ mm^2$

pin support at node ① and roller support at node ③

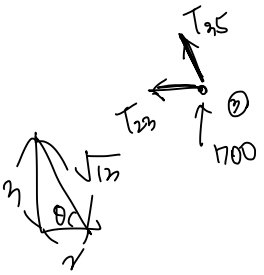
Equilibrium $\Sigma F_x = 0;\ R_{1x} + 800 = 0$ $\therefore R_{1x} = -800\ kN\ (\leftarrow)$

$\Sigma F_y = 0;\ R_{1y} + R_{3y} - 400 - 400 = 0 \Rightarrow R_{1y} + R_{3y} = 800$ ..... ①

$\Sigma M_1 = 0;\ R_{3y} \cdot 8000 - 400 \cdot 2000 - 400 \cdot 6000 - 800 \cdot 3000 = 0$ $\therefore R_{3y} = 1700\ kN\ (\uparrow)$

with Eq ①, $R_{1y} = 800 - 1700 = 100\ kN\ (\uparrow)$

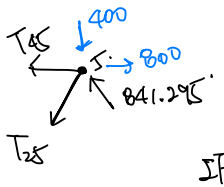
At node 3



$\Sigma F_y = 0;\ T_{35} \cdot \frac{3}{\sqrt{13}} + 1700 = 0$ $\therefore T_{35} = -841.295\ kN$ (compression) : element 7

$\Sigma F_x = 0;\ -T_{35} \cdot \frac{2}{\sqrt{13}} - T_{23} = 0$ $\therefore T_{23} = 466.667\ kN$ (Tension) : element 2

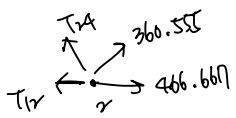
At node 5



$\Sigma F_y = 0;\ 841.295 \cdot \frac{3}{\sqrt{13}} - 400 - T_{25} \cdot \frac{3}{\sqrt{13}} = 0$ $\therefore T_{25} = 360.555\ kN$ (Tension): element 6

$\Sigma F_x = 0;\ -841.295 \cdot \frac{2}{\sqrt{13}} - T_{25} \cdot \frac{2}{\sqrt{13}} + 800 - T_{45} = 0$ $\therefore T_{45} = 133.333\ kN$ (Tension) : element 3

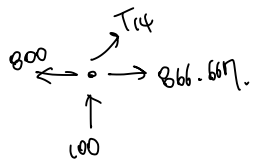
At node 2



$\Sigma F_y = 0;\ T_{24} \cdot \frac{3}{\sqrt{13}} + 360.555 \frac{3}{\sqrt{13}} = 0$ $\therefore T_{24} = -360.555\ kN$ (compression) : element 5.

$\Sigma F_x = 0;\ 360.555 \cdot \frac{2}{\sqrt{13}} + 466.667 - T_{24} \cdot \frac{2}{\sqrt{13}} - T_{12} = 0$ $\therefore T_{12} = 866.667\ kN$ (Tension) : element 1

At node 1



$\Sigma F_y = 0;\ 100 + T_{14} \cdot \frac{3}{\sqrt{13}} = 0$ $\therefore T_{14} = -120.185\ kN$ (compression) : element 4